Experiment - 14

Examine network Address translation (NAT) using cisco packet tracer

Aim:

To configure and examine static NAT, Dynamic NAT, and PAT on cisco router using packet tracer

Procedure

1) create topology: internal PCs -> switch -> router -> External server

2) Assign private IPs to PCs: 192.168.1.10-12

3) assign public IP to the router

4) config router inside/outside interfaces

5) Static NAT: ip nat inside source static

6) Define ACL for dynamic nat

7) create nat pool: ip nat pool MyPool

8) Dynamic nat: ip nat inside source list 1 pool mypool

9) PAT: ip nat inside source list 1 interface

commands / code

```
router (config)
router (config-if)
router (config-if) # ip nat
router (config-if) # no shutdown
router (config) # interface serial
router (config-if) # ip nat outside
router (config-if) # no shutdown
!-- static nat --
router (config) # access list
router (config) # ip nat inside
!-- verification --
router # show ip nat translations
router # show ip nat statistics
router # debug ip nat
```

10) verify with: show ip nat translations
show ip nat statistics

11) Test by ping external server

Result:

NAT was successfully configured on the router. Static NAT, Dynamic NAT and PAT were implemented and verified internal hosts were able to communicate